Grade 11 Chemistry Unit 2 Chemical Bonding

2.1 Introduction

At the end of this section, you will be able to:

- define chemical bonding
- explain why atoms form chemical bonds
- illustrate chemical bonding using the octet rule
- describe the types of chemical bonding and the mechanisms of the bonding process.

2.2 Ionic Bonds

At the end of this section, you will be able to:

- define ionic bonding
- use Lewis electron dot symbols for main group elements
- describe ionic bonding using Lewis electron dot symbols
- list the favourable conditions for the formation of ionic bonds
- explain the formation of ionic bonding
- give examples of ionic compounds
- define lattice energy
- calculate lattice energy of ionic crystals from given data using the Born Haber cycle
- discuss the exceptions to the octet rule
- describe the properties of ionic bonding
- carry out an activity to demonstrate the effect of electricity on ionic compounds (PbI2 and NaCl)
- carry out an investigation into the melting point and solubility of some
- ionic compounds (NaCl and CuCl2).

2.3 Covalent Bonds and Molecular Geometry

At the end of this section, you will be able to:

- define covalent bonding
- explain the formation of covalent bonding using examples
- draw Lewis structures or electron dot formulas of some covalent molecules
- illustrate the formation of coordinate covalent bonding using examples
- draw resonance structures of some covalent molecules and polyatomic ions
- discuss the exceptions to the octet rule in covalent bonding
- distinguish between polar and non-polar covalent molecules
- describe the properties of covalent molecules
- carryout an activity to investigate the effects of heat, electricity and some solvents on covalent compounds (naphthalene, graphite, iodine and ethanol)
- describe the valence shell electron pair repulsion theory (VSEPR)
- distinguish between the bonding pairs and nonbonding pairs of electrons
- describe how electron pair arrangements and shapes of molecules can be predicted from the number of electron pairs explain why double bonds and lone pairs cause deviations from ideal bond angles
- explain the term dipole moment with the help of a diagram
- describe the relationship between dipole moment and molecular geometry
- describe how bond polarities and molecular shapes combine to give molecular polarity
- predict the geometrical shapes of some simple molecules on the bases of hybridization and the nature of electron pairs
- construct models to represent shapes of some simple molecules
- define intermolecular forces
- name the different types of intermolecular forces
- explain dipole-dipole interactions
- give examples of dipole-dipole interaction
- define hydrogen bonding
- explain the effect of hydrogen bond on the properties of substances
- give reasons why hydrogen bonding is stronger than ordinary dipole-dipole interactions
- explain dispersion (London) forces
- give examples of dispersion forces
- predict the strength of intermolecular forces for a given pair of molecules.

2.4 Metallic Bonding

At the end of this section, you will be able to:

- explain how a metallic bond is formed
- explain the properties of metals related to the concept of bonding, and investigate the conductivity, malleability and ductility of some metals and non-metals (Al, Cu, Fe, Sn, Zn, S, C charcoal, C graphite and Si).

2.5 Chemical Bonding Theories

At the end of this section, you will be able to:

- name two chemical bond theories
- explain the valence bond theory
- distinguish the Lewis model and the valence bond model
- discuss the overlapping of orbitals in covalent bond formation
- explain hybridization
- show the process of hybridization involved in some covalent molecules
- draw hybridization diagrams for the formation of sp, sp2, sp3, sp3d and sp3d2 hybrids
- suggest the kind of hybrid orbitals on the basis of the electron structure of the central atom
- predict the geometrical shapes of some simple molecules on the basis of hybridization and the nature of electron pairs
- discuss the hybridization involved in compounds containing multiple bonds
- explain bond length and bond strength
- explain molecular orbital theory
- describe molecular orbital using atomic orbitals
- describe bonding and anti-bonding molecular orbitals
- draw molecular orbital energy level diagrams for homonuclear diatomic molecules
- write the electron configuration of simple molecules using the molecular orbital model
- define bond order and determine the bond order of some simple molecules and molecule-ions
- determine the stability of a molecule or an ion using its bond order; and predict magnetic properties of molecules.

2.6 Types of Crystal

At the end of this section, you will be able to:

- define a crystal
- name the four types of crystalline solid and give examples
- mention the types of attractive force that exist within each type of crystalline solid
- describe the properties of each type of crystalline solid
- build a model of sodium chloride crystal structure.